MDCNet: A Multiscale Dual-Domain Collaborative Network for Semantic Segmentation of Remote Sensing Images

Hailong Ma, Songwei Pei, *Member, IEEE*, Bingfeng Liu, Qian Li, *Member, IEEE*, and Shangguang Wang, *Senior Member, IEEE*

***Abstract*—Semantic segmentation is a fundamental task in remote sensing image (RSI) analysis and plays a vital role in applications such as land-use mapping, urban planning, and environmental monitoring. However, RSIs often suffer from severe grayscale variations and unstable interclass variance, which lead to blurred category boundaries and reduced segmentation accuracy. Most existing methods primarily focus on spatial-domain feature modeling; however, they pay insufficient attention to the rich information embedded in the frequency domain, which consequently limits segmentation performance in complex scenarios. To address these challenges, this article proposes a multiscale dual-domain collaborative network (MDCNet), which innovatively introduces a spatial-frequency collaborative module (SFCM) to separate high- and low-frequency information using discrete wavelet transform (DWT) and to deeply fuse spatial- and frequency-domain features through a cross-attention mechanism. In addition, a multiscale gated perception bridge (MGB) is designed to promote efficient interaction among multiscale features, and a multidimensional feature refinement head (MFRH) is introduced to reduce the semantic gap between deep and shallow features, thereby improving overall segmentation performance. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets demonstrate that MDCNet consistently outperforms existing methods in segmentation accuracy and robustness.**

***Index Terms*—Remote sensing, semantic segmentation, Transformer, wavelet transform.**

I. Introduction

With the rapid development of remote sensing technology, high-resolution remote sensing images (RSIs) provide a solid data foundation for applications including land use [1], urban planning [2], and environmental protection [3]. As a key task in RSI interpretation, semantic segmentation assigns semantic categories to each pixel. However, with the continuous enhancement of image resolution and content complexity [4], traditional feature engineering-based machine learning methods [5] exhibit obvious limitations when dealing with complex scenarios, making it difficult to meet growing demands for accuracy and robustness.

In recent years, convolutional neural networks (CNNs), with their powerful hierarchical feature extraction capabilities, have greatly advanced the field of RSI segmentation. FCN [6] was the first to achieve end-to-end pixel-level segmentation; however, due to its relatively simple decoder structure, it is prone to the loss of multiscale information. U-Net [7] significantly improved segmentation accuracy by introducing skip connections and multiscale feature fusion, becoming the dominant architecture in this field. Nevertheless, these methods are constrained by limited receptive fields [8], making it difficult to effectively model long-range dependencies. To address this issue, attention mechanisms such as DANet [9], SE [10], and CBAM [11] have been introduced to enhance global contextual correlations and channel selectivity. However, these approaches still fundamentally rely on the local operations of CNNs, rendering them insufficient for capturing comprehensive global information in complex RSI scenarios.

The introduction of Transformers has brought stronger global modeling capabilities to semantic segmentation of RSIs. Vision Transformer (ViT) [12] directly captures long-range dependencies through multihead self-attention mechanisms, significantly enhancing the model's understanding of global semantics. However, its high computational cost limits its application to high-resolution RSIs. Pyramid ViT (PVT) [13] and Swin Transformer [14] effectively alleviate this issue through pyramid structures and window-based self-attention mechanisms, enabling efficient dense prediction.

Despite these notable advances, semantic segmentation of high-resolution RSIs still faces many challenges in real-world applications. In high-density urban scenes, small objects such as vehicles, street facilities, or narrow roads often occupy only a few pixels, making them easily overwhelmed by surrounding context or misclassified as background. In addition, imaging conditions can change drastically over time and with weather. Seasonal transitions alter vegetation phenology and surface reflectance properties, while atmospheric factors such as haze, cloud cover, or low illumination introduce severe grayscale

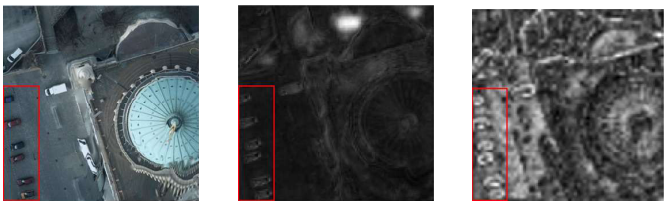
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The authors are with the School of Computer Science (National Pilot Software Engineering School), Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: hailongma@bupt.edu.cn; peisongwei@bupt.edu.cn; bfliu@bupt.edu.cn; li.qian@bupt.edu.cn; sgwang@bupt.edu.cn).

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(a) (b)

Fig. 1. Visualization of RSI features. (a) Feature map extracted by the Transformer. (b) High-frequency feature map extracted using the Haar transform.

inconsistency and reduced contrast, which further aggravate interclass confusion.

However, as pointed out by Zhang et al. [15], Transformers primarily learn low-frequency components in the spatial domain and exhibit insufficient modeling capability for high-frequency details. This limitation is particularly pronounced in fine-grained segmentation tasks, often resulting in blurred object boundaries and misclassification. In remote sensing imaging, complex illumination and atmospheric conditions, such as varying solar illumination, cloud shadows, and atmospheric scattering, frequently cause severe grayscale variations, significantly altering the absolute pixel intensities of the same ground objects. Nevertheless, the intrinsic physical structures and material boundaries of these objects remain relatively invariant under such illumination changes. Edge and texture cues are originally implicit in pixel gradients, but through frequency-domain decomposition, they can be transformed into clearer spectral responses, enabling the network to maintain consistent and stable feature representations even under large grayscale fluctuations. In addition, to address the instability of interclass differences, high-frequency components can serve as "structural priors," guiding spatial features to focus more on subtle texture differences and contour discontinuities, thereby promoting more accurate pixel-level classification.

The feature map extracted by the Transformer, as shown in Fig. 1(a), exhibits a pronounced low-frequency bias, which tends to smooth the boundaries of small-scale targets such as vehicles and causes the loss of fine details. On the contrary, the high-frequency feature map obtained through the Haar wavelet transform, as shown in Fig. 1(b), better preserves spatial details and contour gradients, making the boundaries of small objects clearer and more precise. These results demonstrate that incorporating frequency-domain information into spatial features can effectively alleviate the boundary blurring problem in small-object segmentation.

As a classic frequency-domain analysis tool, the wavelet transform has attracted considerable attention in the field of image processing [16], [17] in recent years. The Haar wavelet transform can effectively decompose an image into high-frequency and low-frequency components, thereby enabling the extraction of edge textures and global structural information, respectively. For instance, FcaNet [17] and FsaNet [15] introduce frequency channel attention mechanisms, which effectively preserve fine textures and improve segmentation accuracy. However, most existing methods simply replace spatial-domain features with frequency-domain features, lacking deep integration and collaborative modeling between the two. As a result, the complementary advantages of spatial- and frequency-domain features have not been fully exploited.

To address the semantic segmentation of high-resolution RSIs, this article proposes a multiscale dual-domain collaborative network (MDCNet). Although recent work such as DOCNet [18] also employs the term "dual-domain," it defines the dual domains as the "spatial and semantic domains," focusing on leveraging class-aware priors to optimize feature distributions and alleviate intraclass variance. On the contrary, MDCNet defines the dual domains as the "spatial and frequency domains." To this end, we design a spatial-frequency collaborative module (SFCM) based on the discrete wavelet transform (DWT), which decouples features into high- and low-frequency components and injects spectral responses into the network as structural priors to enhance spatial-domain representations. Furthermore, to strengthen multiscale feature representation, we introduce a multiscale gated perception bridge (MGB), which replaces the standard U-Net skip connections with a gating mechanism to enable adaptive cross-scale fusion. This effectively reduces semantic gaps and collaborates with the SFCM to preserve specific fine-grained details. Finally, a multidimensional feature refinement head (MFRH) is employed as the segmentation head of the decoder to refine the aggregated features, thereby improving both segmentation accuracy and robustness. In summary, the main contributions of this article are as follows.

1) We propose an MDCNet for semantic segmentation of RSIs. By deeply integrating frequency-domain and spatial-domain features, the network is able to extract salient features that effectively characterize intraclass consistency and interclass differences. Consequently, MDCNet can robustly address common challenges in RSIs, such as drastic grayscale variations and unstable interclass variance.
2) We innovatively design an SFCM and an MGB to achieve efficient fusion of dual-domain features and enhance the representational capacity of spatial features. With the scale-aware gating mechanism, our method can efficiently model long-range dependencies among multiscale features, further improving the ability of the SFCM to capture both detailed and global information. Meanwhile, the proposed MFRH adaptively fuses features at different levels, enabling efficient feature aggregation and further improving segmentation accuracy.
3) Experimental results on multiple public remote sensing datasets, including ISPRS Vaihingen [19], ISPRS Potsdam [19], and LoveDA [20], fully validate the effectiveness and robustness of the proposed MDCNet. Compared with existing mainstream methods, MDCNet demonstrates superior segmentation performance and broad application potential.

II. Related Work

A. Multiscale Feature Aggregation

Objects in RSIs exhibit significant scale variations, which places high demands on the multiscale feature extraction

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capability of segmentation networks. Traditional fixed-size convolution kernels are limited in their ability to represent features across different scales. To address this, researchers have proposed various multiscale feature extraction architectures. For example, MACU-Net [21] employs asymmetric convolutions and multiscale skip connections to achieve adaptive feature fusion. ResUNet-a [22] integrates residual connections and atrous convolutions, thereby enhancing the learning of multiscale features. HFGNet [23] designs parallel multisize convolution modules to extract multiscale representations and facilitate efficient feature interaction. DeepLabV3+ [24] introduces the ASPP module and atrous convolutions with different dilation rates to enable effective multiscale feature fusion. Although these methods enhance feature extraction capabilities, their ability to model large receptive fields remains limited, and enlarging the convolution kernel size typically leads to increased parameter counts and computational costs [25].

To further enhance the fusion and utilization of multiscale features, researchers have proposed various strategies to strengthen feature interaction. CFNet [26] enriches multiscale representations by fusing features from adjacent layers. SSAtNet [27] adopts a parallel structure inspired by ASPP and employs a stepwise aggregation mechanism to integrate features with different receptive fields. Although these methods have made progress in multiscale feature extraction and fusion, the interaction between multiscale features still needs to be strengthened, and the potential of networks to exploit multiscale information has yet to be fully realized.

B. Semantic Segmentation for RSIs

Relying solely on convolution-based feature extraction often leads to overly localized features, making global contextual information critically important. Self-attention mechanisms have been widely adopted due to their ability to effectively model long-range dependencies. MAResU-Net [28] introduces a linear attention mechanism into the skip connections of U-Net, enhancing the modeling of long-range dependencies and improving segmentation accuracy. The block-level attention module proposed in LANet [29] preserves local details during multilevel feature fusion, thereby improving the recognition of small objects and boundaries. However, these methods still aggregate global information based on local features extracted by CNNs, making it difficult to meet the demands of high-resolution RSIs for robust global perception.

Transformer models explicitly model global dependencies through multihead self-attention mechanisms, establishing a new paradigm for semantic segmentation. However, ViT suffers from low computational efficiency when processing large-scale images. The Swin Transformer addresses this issue by introducing window-based attention; nevertheless, approximating global self-attention still requires stacking numerous modules, which increases model complexity. To address these issues, Blaga and Nedevschi [30] proposed a Transformer-based U-Net variant incorporating focal-axial attention, which significantly enhances segmentation performance for high-resolution RSIs. GLOTS [31] adopts a unified Transformer encoder–decoder architecture combined with global–local attention mechanisms, effectively capturing multiscale contextual information and improving model generalization. CLCFormer [32] integrates CNNs with Transformers to balance fine-grained spatial details and long-range dependencies, thereby improving segmentation accuracy. Wu et al. [33] propose Tiff-SegFormer, which fuses spectral and thermal infrared features via dual encoders for multimodal collaboration. AerialFormer [34] aggregates multiscale Transformer features using a multidilation convolutional decoder to improve segmentation of small and densely distributed objects. GDGNet [35] mitigates low-level feature entanglement and enhances spatial detail extraction through gradient decoupling guidance and global–local context fusion. CG-FCLNet [36] integrates multiscale context via class-guided feature collaboration to improve boundary accuracy and detail preservation. D2LS [37] explicitly models class semantics using dynamic dictionary learning with multistage cross-attention and contrastive constraints, enhancing discriminability among morphologically similar classes.

In recent years, Mamba has demonstrated its effectiveness across a variety of visual tasks. For example, Samba [38] is the first to introduce the Mamba state space model into RSI segmentation, efficiently capturing global semantics through an encoder–decoder architecture and significantly improving segmentation performance on high-resolution images. RSMamba [39] is based on a state space model and introduces a dynamic multipath activation mechanism to effectively model global dependencies in RSIs, significantly improving scene classification performance. RS3Mamba [40] adopts a dual-branch architecture, in which a Mamba-based auxiliary branch provides global context to the CNN-based main branch, thereby enhancing category consistency and structural awareness. RTMamba [41] introduces Mamba modules into the deeper stages of the encoder for long-range modeling, and leverages its linear complexity to meet real-time requirements. UNetMamba [42] proposes a lightweight U-shaped architecture based on Mamba, combining an efficient decoder and a local supervision module to achieve high accuracy in RSI segmentation while maintaining low computational cost. DiffMamba [43] combines diffusion models and state space models, effectively fusing local and global features through semantic diffusion alignment and coordinate-aware Mamba modules to improve RSI segmentation accuracy.

Built upon state-space models, Mamba achieves linear complexity of $O(N)$ through recursion. It compresses historical information into a fixed-size hidden state, decoupling computational cost from sequence length. Its "selective" mechanism makes the parameters dependent on the current input, enabling content-aware gating that effectively filters noise and propagates salient information over long sequences. From a frequency-domain perspective, CNNs tend to capture local high-frequency details, whereas Transformers are more inclined to model global low-frequency structures; Mamba dynamically balances high- and low-frequency components. Its input-dependent discretization step Δ acts as a frequency regulator: smaller values approximate integration to preserve low-frequency trends, while larger values trigger rapid updates

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to capture high-frequency abrupt changes, thereby adaptively shifting spectral attention according to the input.

C. Frequency-Domain Learning

In recent years, frequency information has demonstrated unique advantages in RSI analysis. By employing methods such as the DWT [44] and the Fourier transform (FFT) [45], models can decompose images into high- and low-frequency components, enabling precise multiscale feature extraction. Frequency-domain features play a crucial role in discriminating complex ground objects. For example, FHRB-Net [46] introduces a frequency-domain heterogeneous modulation fusion module for change detection, which decouples bitemporal features into high- and low-frequency components via the Fourier transform and employs heterogeneous modulation strategies to enhance change information while suppressing style interference. GLFRNet [47] leverages the Fourier transform to jointly reconstruct low-frequency structural information from DSM data and high-frequency texture details from optical images, achieving effective cross-modal feature fusion through frequency-domain decomposition and phase-guided compensation. FreMIM [48] leverages the Fourier transform to formulate a frequency-domain masked modeling task, enhancing the multiscale representation capability for segmentation by decoupling and recovering high- and low-frequency components. However, the Fourier transform lacks spatial localization capability, making it difficult to effectively capture abrupt local details in RSIs. Similarly, although Dunn and Higgins [49] optimized Gabor filters to map frequency-domain textures into step-like response signals in the spatial domain, the nonorthogonality and computational redundancy of the Gabor transform limit its applicability in efficient neural networks.

On the contrary, the Haar wavelet transform [50] is better suited for high-resolution RSIs. The Fourier transform and Gabor transform approximate signals using continuous sinusoidal basis functions, whereas boundaries of remote sensing targets typically exhibit instantaneous jumps in pixel intensity. Representing such sharp edges, therefore, often requires superimposing a large number of high-frequency components, leading to significant computational redundancy. On the contrary, the Haar wavelet employs discontinuous step-shaped basis functions, and its computation is primarily based on addition and subtraction operations, making it more efficient. Moreover, the form of its basis functions aligns more closely with the step-like nature of object edges, enabling it to capture pixel discontinuities in a sparser and more accurate manner. FreDNet [51] is a frequency-guided denoising network that treats high-frequency signals as semantic cues via the DWT and adaptively enhances or suppresses them through structured frequency modulation, thereby improving boundary preservation in RSI segmentation.

III. Method

To improve the accuracy and robustness of semantic segmentation for RSIs, we propose the MDCNet. First, MDCNet employs a multiscale gated perception bridge (MGB) to replace traditional skip connections. Through a gating mechanism, MGB effectively fuses multiscale features from the encoder, providing refined and enriched representations for the decoder. Second, an SFCM is designed within the decoder to deeply integrate spatial- and frequency-domain features. By exploiting the complementarity of dual-domain information, SFCM strengthens target consistency and category discrimination in complex scenes. Finally, an MFRH performs weighted fusion and fine-grained optimization on the output features, further enhancing overall segmentation accuracy and detail preservation.

A. Multiscale Gated Perception Bridge

Although the Swin Transformer efficiently extracts multiscale features through a sliding window mechanism, features from a single stage often struggle to simultaneously capture fine-grained local details and large-scale global semantic information. This limitation is particularly pronounced in RSIs, where phenomena such as unstable interclass variance impose higher demands on the model's generalization capability. To address this issue, we design an MGB to supplement and fuse multiscale features from all stages of the encoder, thereby enabling efficient integrated learning of cross-scale information.

As shown in Fig. 2, the MGB takes as input the output features F_e^i from the four stages of the Swin Transformer encoder, where $i \in \{1, 2, 3, 4\}$. Each F_e^i is first downsampled by a factor of two using max pooling, reducing its spatial size from (H_i, W_i) to $(H_i/2, W_i/2)$. Subsequently, all downsampled features are reshaped to share a unified channel dimension and concatenated along the flattened spatial dimension to form the fused feature F_c, whose total sequence length is $N = \sum_{i=1}^{4} H_i W_i/4$. The fused feature F_c then passes sequentially through one linear attention module followed by three condensed attention modules to model cross-scale dependencies. Meanwhile, the introduced scale-aware gating mechanism adaptively reweights features from different scales, thereby alleviating semantic discrepancies during the fusion process.

1) Condensed Attention: As shown in Fig. 2(a), condensed attention is a spatial dimension reduction attention mechanism, which is essentially equivalent to the traditional dot-product attention module [52] but incorporates a scale compression module. This mechanism first maps the input feature Y_c into a query vector $Q \in \mathbb{R}^{N \times d_k}$. When generating the key (K) and value (V), a scale compression module is introduced to reduce redundant information in high-resolution features, thereby effectively lowering computational complexity and improving the processing efficiency of high-resolution feature maps. Its mathematical expression is as follows:

$$\begin{aligned} K_r &= \text{Reshape}\left(K, \left(\frac{N}{r}, rC\right)\right) \text{Proj}\,(rC, r) \\ V_r &= \text{Reshape}\left(V, \left(\frac{N}{r}, rC\right)\right) \text{Proj}\,(rC, r). \end{aligned} \tag{1}$$

Here, K_r denotes the new key after scale compression. By applying scale compression with a ratio of r, the sequence length is reduced from N to N/r, while the number of channels

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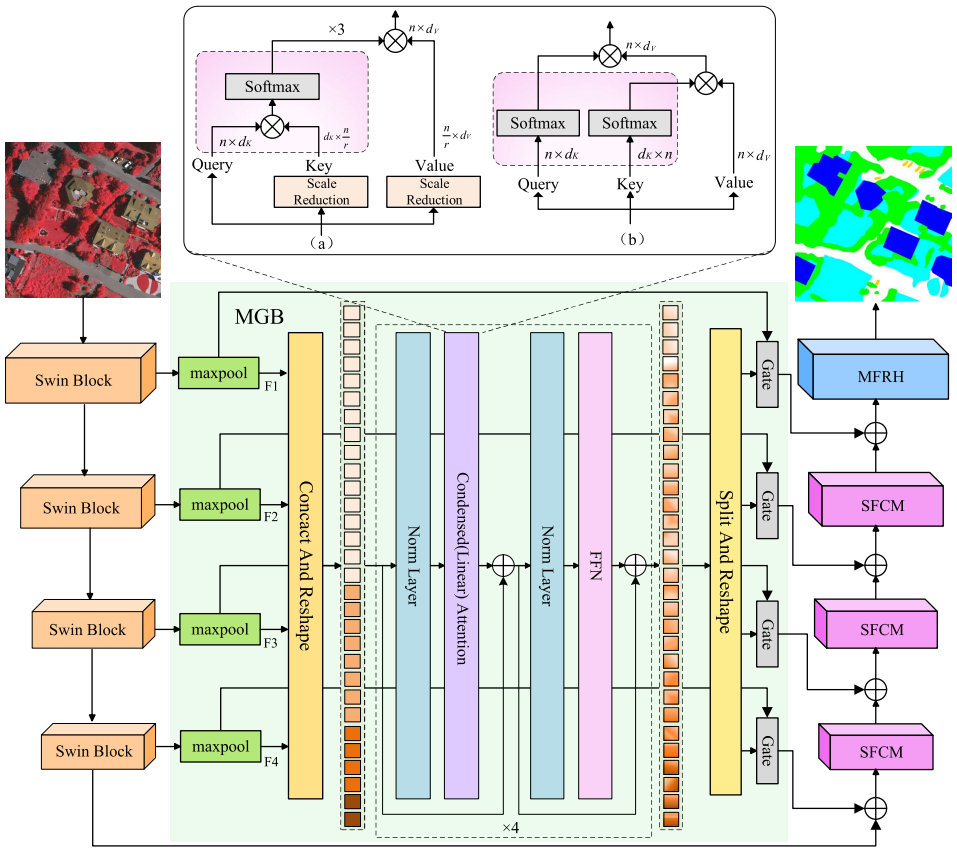


Fig. 2. Proposed MDCNet framework consists of the MGB, the SFCM, and the MFRH.

increases from C to rC. Subsequently, a linear projection layer (Proj) restores the channel depth from rC to r to ensure consistency in the subsequent processing steps. V_r is generated in the same way as K_r.

Next, condensed attention computes the similarity between Q and the scale-compressed K_r to obtain the attention weights for V_r, thereby effectively capturing cross-scale relational information between different stages and further reducing overall computational complexity. The computation is formulated as follows:

$$\text{Atten}_C\left(Q, K_r, V_r\right) = \text{Softmax}\left(\frac{QK_r^T}{\sqrt{d_k}}\right)V_r. \tag{2}$$

2) Linear Attention: To reduce computational complexity, we introduce linear attention into the MGB, as illustrated in Fig. 2(b). Given the input feature Y_c, it is first projected into a query $Q \in \mathbb{R}^{N\times d_k}$, a key $K \in \mathbb{R}^{N\times d_k}$, and a value $V \in \mathbb{R}^{N\times d_v}$. By reordering the computation of standard dot-product attention from $(QK^\top)V$ to $Q(K^\top V)$, and leveraging the associativity of matrix multiplication, the computational complexity is reduced from $O(N^2)$ to $O(N)$ [53].

Here, $\text{Softmax}_{\text{col}}(\cdot)$ and $\text{Softmax}_{\text{row}}(\cdot)$ denote softmax normalization applied along the column and row dimensions of a matrix, respectively. For an arbitrary matrix A, they are defined as $\text{Softmax}_{\text{col}}(A)_{i,j} = \exp(A_{i,j})/\sum_k \exp(A_{k,j})$ and $\text{Softmax}_{\text{row}}(A)_{i,j} = \exp(A_{i,j})/\sum_k \exp(A_{i,k})$. Finally, the output of the linear attention is computed as follows:

$$LA = \text{Softmax}_{\text{col}}\left(\frac{K}{\sqrt{N}}\right)^T \times V$$
$$\text{Atten}_L\left(Q, K, V\right) = \text{Softmax}_{\text{row}}\left(\frac{Q}{\sqrt{N}}\right) \times LA. \tag{3}$$

3) Scale-Aware Gate: The output of the scale-aware gate is utilized for the final mask prediction in the decoder, aiming to mitigate potential semantic discrepancies during multiscale attention fusion and enhance the model's adaptability and semantic consistency in multiscale information fusion. In particular, we split and reshape the fused feature F_c according to the original scales of the encoder stages to obtain F_c^i, ensuring that it is spatially and channelwise consistent with the encoder feature F_e^i. On this basis, the weights for cross-scale interaction are dynamically calculated in combination with F_e^i. The calculation formula is as follows:

$$F_o^i = \text{sigmoid}\left(F_e^i W_1\right) \otimes \left(F_c^i W_2\right) + F_e^i W_3. \tag{4}$$

Here, $i \in 1, 2, 3, 4$ denotes the feature level, and $\otimes$ represents elementwise multiplication. W_1, W_2, and W_3 are learnable parameter matrices used for feature transformation and gate generation. To ensure gradient flow and convergence stability during the initial training phase, we adopt the Xavier initialization [54] strategy for these weight parameters.

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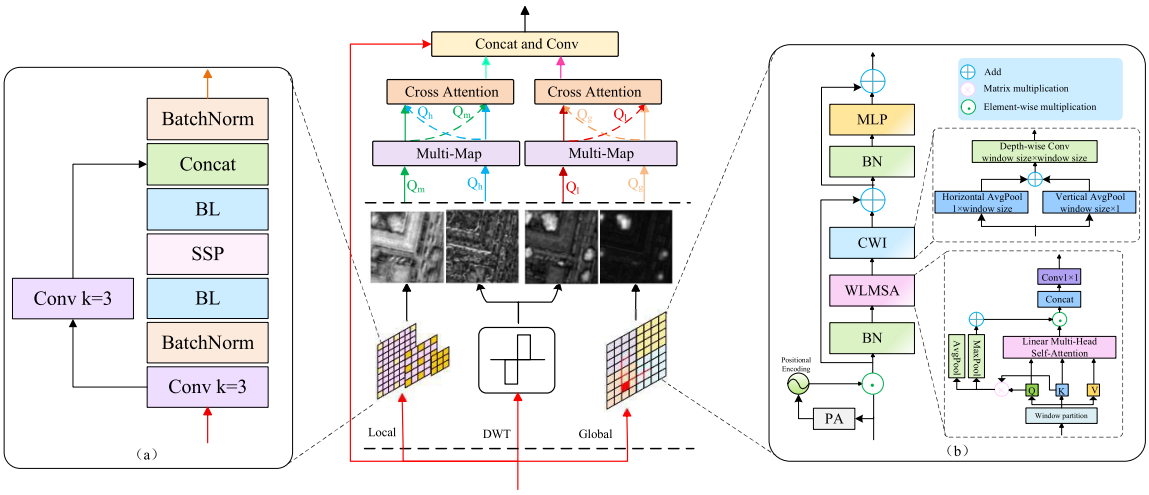


Fig. 3. Schematic of the SFCM module structure.

B. Space–Frequency Domain Collaborative Module

The structure of SFCM is shown in Fig. 3. It adopts a parallel three-branch architecture, comprising a DWT branch, an improved Swin Transformer global branch, and a multiscale convolutional local branch. These branches generate four complementary feature maps in the frequency and spatial domains: high-frequency features F_H, low-frequency features F_L, local features F_m, and global features F_g. Subsequently, a cross-attention mechanism is employed to align and fuse the dual-domain features, thereby enhancing the model's ability to distinguish different categories in regions with varying gray levels.

1) DWT Branch: DWT is an efficient frequency-domain signal processing method [44]. By cascading high-pass and low-pass filters, it separates high- and low-frequency information across different subbands at a relatively low computational cost, thereby enriching feature representation.

In RSI processing tasks, high-frequency components correspond to regions where pixel values change rapidly. For example, the boundary between a building and the ground often exhibits an "instantaneous step" in brightness. Such characteristics enable high-frequency information to accurately capture object contours and fine texture details. On the contrary, low-frequency components correspond to smooth regions where pixel values vary slowly, such as large lakes or flat farmlands, and thus primarily reflect the global structure and spatial distribution of objects.

Significant grayscale variations in RSIs are typically caused by uneven illumination or atmospheric interference, such as cloud shadows or haze. These factors mainly affect the overall image brightness, which corresponds to the low-frequency component. On the other hand, the high-frequency components extracted by the Haar wavelet capture relative gradient changes between neighboring pixels. These gradients reflect intrinsic geometric structures and material differences and are highly robust to grayscale fluctuations. Therefore, explicitly decoupling and enhancing high-frequency components allows the model to effectively resist unstable grayscale variations, focus on illumination-invariant physical boundaries, and significantly improve feature reliability in complex environments.

Let k denote the recursion level of decomposition. The input to the kth level DWT is the low-frequency feature map F_{k-1}, and the DWT outputs four subband components. Taking $k = 1$ as an example, the input feature map is denoted as $X \in \mathbb{R}^{H \times W \times C}$. DWT uses nonoverlapping local blocks of size 2×2 as the basic unit: each output position (i, j) corresponds to four pixels $(2i-1, 2j-1)$, $(2i-1, 2j)$, $(2i, 2j-1)$, and $(2i, 2j)$ in the feature map. Through fixed linear combinations with addition/subtraction weights and downsampling, X can be decomposed into three high-frequency subfeatures $(F^k_{LH}, F^k_{HL}, F^k_{HH})$ and one low-frequency subfeature $F^k_{LL}(i, j)$. In addition, we use a preset hyperparameter k and select the four subband components at level $k = 3$ as the final outputs $(F_{LL}, F_{LH}, F_{HL}, F_{HH})$. The specific calculation is as follows:

$$\begin{cases} F_{LL} & = F_{k-1}(2i-1, 2j-1) + F_{k-1}(2i-1, 2j) \\ & \quad + F_{k-1}(2i, 2j-1) + F_{k-1}(2i, 2j) \\ F_{LH} & = -F_{k-1}(2i-1, 2j-1) - F_{k-1}(2i-1, 2j) \\ & \quad + F_{k-1}(2i, 2j-1) + F_{k-1}(2i, 2j) \\ F_{HL} & = -F_{k-1}(2i-1, 2j-1) + F_{k-1}(2i-1, 2j) \\ & \quad - F_{k-1}(2i, 2j-1) + F_{k-1}(2i, 2j) \\ F_{HH} & = F_{k-1}(2i-1, 2j-1) - F_{k-1}(2i-1, 2j) \\ & \quad - F_{k-1}(2i, 2j-1) + F_{k-1}(2i, 2j). \end{cases} \tag{5}$$

Here, the range of i is from 1 to $H/2$, and the range of j is from 1 to $W/2$. F_{LL} is the low-frequency component, reflecting the overall structural information of the feature map; F_{LH} and F_{HL} correspond to the high-frequency components in the horizontal and vertical directions, respectively, mainly used to capture edge and texture variations; F_{HH} focuses on high-frequency details in the diagonal direction.

Subsequently, the three high-frequency subfeatures produced by DWT are concatenated to form a high-frequency signal, which is then fed into a nonlinear mapping operation to accomplish feature reconstruction and channel compression. In this way, redundant information can be filtered out,

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yielding the final high-frequency and low-frequency features. The formulation is given as follows:

$$F_l, F_h = \psi\left(\delta_{1\times1}\left(F_{LL}\right)\right), \psi\left(\delta_{1\times1}\left(\text{Cat}\left(F_{LH}, F_{HL}, F_{HH}\right)\right)\right). \quad (6)$$

Here, F_l and F_h denote the low-frequency and high-frequency features, respectively, both with dimensions $(C, H/2, W/2)$, $\psi(\cdot)$ denotes batch normalization, $\delta_{k\times k}(\cdot)$ denotes a convolution operation with kernel size $k \times k$, and $Cat(\cdot)$ denotes channel concatenation.

2) Improved Swin Transformer Global Branch: While Swin Transformer [14] improves efficiency via a sliding window mechanism, increasing window sizes still incurs high computational burdens [55]. Therefore, we introduce linear attention, reducing the complexity of window-based multihead self-attention (W-MSA) from $O(N^2)$ to $O(N)$, where N is the window sequence length. Concurrently, we replace the complex SW-Trans module with more efficient stripe pooling (SP) and large kernel convolutions to achieve efficient contextual interaction between windows.

As shown in Fig. 3(b), the global branch adopts the pixel attention (PA) method from ResT [56] for position encoding. PA obtains pixel-level weights through a 3×3 depthwise separable convolution, embeds positional information via elementwise multiplication, and scales the result using the Sigmoid function. The specific formula is as follows:

$$X = \text{PA}\left(X_{\text{in}}\right) = X_{\text{in}} \odot \sigma\left(\text{DWConv}_{3\times3}\left(X_{\text{in}}\right)\right). \quad (7)$$

Here, X_{in} denotes the input features, $\odot$ denotes elementwise multiplication, σ denotes the Sigmoid function, and $\text{DWConv}_{3\times3}$ denotes a 3×3 depthwise separable convolution.

After position encoding is completed, the input feature map is divided into a series of nonoverlapping windows. For the ith attention head in the nth window, its self-attention inputs Q_n^i, K_n^i, and V_n^i are obtained through three sets of linear layers, as follows:

$$Q_n^i = X_n W_Q^i, \quad K_n^i = X_n W_K^i, \quad V_n^i = X_n W_V^i. \quad (8)$$

Here, $W_Q^i, W_K^i \in \mathbb{R}^{C\times(C/4I)}$, $W_V^i \in \mathbb{R}^{C\times(C/I)}$, $Q_n^i, K_n^i \in \mathbb{R}^{S^2\times(C/4I)}$, $V_n^i \in \mathbb{R}^{S^2\times(C/I)}$, and $X_n \in \mathbb{R}^{S^2\times C}$. C represents the number of channels, S represents the window size, and I represents the number of self-attention heads.

Next, linear attention is computed for each attention head, as shown in (3). It should be noted that linear attention only learns contextual relationships of positional information from the spatial dimension. To supplement the dependencies between channels, channel attention is further introduced, enabling the global branch to capture rich information in both spatial and channel dimensions. The calculation method for channel attention is as follows:

$$\begin{aligned}\text{CA}\left(Q_i^n, K_i^n\right) = &\text{maxpool}\left(\rho_{\text{row}}\left(Q_i^n\right)^T \odot \rho_{\text{col}}\left(K_i^n\right)\right)\\ &\oplus \text{avgpool}\left(\rho_{\text{row}}\left(Q_i^n\right)^T \odot \rho_{\text{col}}\left(K_i^n\right)\right). \quad (9)\end{aligned}$$

Finally, linear multihead self-attention within the window (WLMSA) is implemented, and its expression is as follows:

$$\begin{aligned}\text{head}_n^i &= \text{LA}\left(Q_n^i, K_n^i, V_n^i\right) \odot \text{CA}\left(Q_n^i, K_n^i\right)\\ \text{WLMSA}\left(X_n\right) &= \text{Concat}\left(\text{head}_n^1, \ldots, \text{head}_n^I\right) W^o. \quad (10)\end{aligned}$$

The shifted window mechanism of Swin Transformer significantly increases computational complexity in high-resolution scenarios such as RSIs. To address this, this article replaces the SW-Trans module with SP and large-kernel convolution to improve the efficiency of contextual interaction between windows. In particular, the output of WLMSA is first fed into two average pooling layers (with kernel sizes of $1 \times S$ and $S \times 1$), which perform long-range information fusion in the horizontal and vertical directions, respectively, both with a stride of 1. Then, the outputs of these two pooling layers are summed elementwise and further fused spatially through an $S \times S$ depthwise separable convolution.

3) Multiscale Convolutional Local Branch: Adopting pooling windows of different sizes enables the model to effectively capture multiscale features, enhancing the recognition of local details and complex patterns. As shown in Fig. 3(a), for the input feature $x \in \mathbb{R}^{C\times H\times W}$, the local branch employs 5×5, 9×9, and 13×13 max pooling kernels to extract and fuse multiscale local features. The detailed process is as follows:

$$\begin{aligned}f_{\text{BL}}\left(x\right) &= \delta_{3\times3}\left(\delta_{1\times1}\left(x\right)\right)\\ f_{\text{SPP}}\left(x\right) &= \text{Cat}\left(\text{MP}_{5\times5}\left(x\right), \text{MP}_{9\times9}\left(x\right), \text{MP}_{13\times13}\left(x\right), x\right)\\ F_m &= \psi\left(\text{Cat}\left(\delta_{3\times3}\left(\delta_{3\times3}\left(x\right)\right) f_{\text{BL}}\left(f_{\text{SPP}}\left(f_{\text{BL}}\left(\psi\left(\delta_{3\times3}\left(x\right)\right)\right)\right)\right)\right)\right). \quad (11)\end{aligned}$$

Here, $f_{\text{BL}}(\cdot)$ denotes the bottleneck layer, $f_{\text{SPP}}(\cdot)$ denotes the spatial pyramid pooling operation, $\text{MP}_{k\times k}(\cdot)$ denotes the max pooling operation with a kernel size of $k \times k$, and $F_m \in \mathbb{R}^{2C\times H/2\times W/2}$ represents the output features of the local branch.

To bridge the semantic gap between frequency and spatial domains, semantic alignment is performed. As shown in Fig. 3, the features are divided into two groups: $\{F_m, F_h\}$ and $\{F_l, F_g\}$. The high-frequency feature F_h captures local variations and boundary cues, while the local feature F_m provides complementary fine-grained spatial information. The low-frequency feature F_l preserves coarse structural information but lacks fine details, and is therefore semantically aligned with the global feature F_g, which provides complementary contextual semantics.

In particular, we utilize multiscale separable convolution (MSC) to process the spatial-domain features F_s and frequency-domain features F_f, respectively. Subsequently, the processed features are projected via 1×1 convolutions to generate Q_1, K_1, V_1 for the spatial branch and Q_2, K_2, V_2 for the frequency branch. For a given input tensor $X \in \mathbb{R}^{C\times H\times W}$, the MSC operation is defined as follows:

$$Q,K,V = \delta_{1\times1}(\text{MSC}_7(\text{LN}(x)) + \text{MSC}_{11}(\text{LN}(x)) + \text{MSC}_{21}(\text{LN}(x))). \quad (12)$$

Here, MSC_k denotes a factorized convolution block with an effective receptive field of size k, which consists of two sequential convolutional layers with kernel sizes of $1 \times k$ and $k \times 1$, respectively. $\text{LN}(\cdot)$ denotes layer normalization, and $\delta_{1\times1}(\cdot)$ represents a 1×1 convolution operation.

For spatial-domain features F_s and frequency-domain features F_f, we adopt a multihead cross-attention mechanism to perform cross-domain semantic alignment. We set the number

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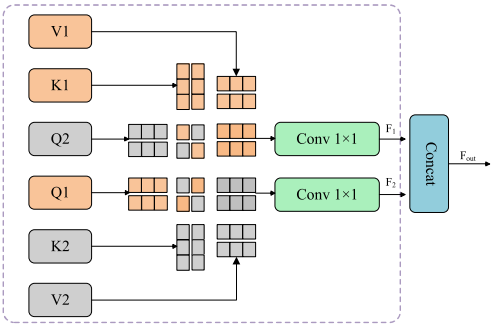


Fig. 4. Cross-attention mechanism is employed to fuse the spatial and frequency domains, facilitating dynamic interaction and adaptive feature selection.

of attention heads to $h = 8$ and the feature dimension of each head to $d_k = C/h$.

Within this mechanism, the quantitative criterion for semantic alignment is implicitly defined by the attention scores between the query and the key. In particular, the model measures the semantic correlation between spatial features and frequency-domain features by computing the scaled dot-product similarity; a larger dot-product value indicates a higher degree of semantic alignment between the two, thereby guiding the network to aggregate the most relevant complementary information from the corresponding domain.

Based on the aforementioned multiscale convolutional mappings, we obtain the projections $\{Q_1, K_1, V_1\}$ from the spatial branch and $\{Q_2, K_2, V_2\}$ from the frequency branch, respectively. As illustrated in Fig. 4, attention maps are generated via cross-computation to enable information fusion between features. The computation is as follows:

$$\begin{aligned} \text{Attn}_1(Q_2, K_1, V_1) &= \text{Softmax}\left(\frac{Q_2 K_1^T}{\sqrt{d_k}}\right) V_1 \\ \text{Attn}_2(Q_1, K_2, V_2) &= \text{Softmax}\left(\frac{Q_1 K_2^T}{\sqrt{d_k}}\right) V_2. \end{aligned} \tag{13}$$

Here, $\sqrt{d_k}$ serves as a scaling factor to regulate the magnitude of the dot products. Subsequently, the two features F_1 and F_2 output by the cross-attention are calculated as follows:

$$F_1 = \delta_{1\times1}(\text{Attn}_1), \quad F_2 = \delta_{1\times1}(\text{Attn}_2). \tag{14}$$

Finally, the two features F_1 and F_2 output by the cross-attention are concatenated along the channel dimension to obtain the aligned output tensor F_{out}

$$F_{\text{out}} = \text{Cat}(F_1, F_2). \tag{15}$$

C. Multidimensional Feature Refinement Head

The shallow features extracted by the first layer of the Swin Transformer primarily preserve basic structural details but lack high-level semantic information. After spatial-frequency feature modeling, the deep features become more discriminative; however, detail loss may occur during frequency transformation and feature aggregation, particularly in high-resolution RSIs. To address this issue, the proposed MFRH module effectively fuses shallow and deep features to enhance segmentation accuracy. As shown in Fig. 5, MFRH first performs a weighted fusion to obtain the input feature $F \in \mathbb{R}^{C\times H\times W}$, and then enhances channel, multiscale, and spatial feature representations through three parallel branches.

The channel branch applies global average pooling to the input feature to generate a channel descriptor $F_p \in \mathbb{R}^{C\times1\times1}$. Subsequently, an MLP followed by a Sigmoid function generates the channel weight distribution, which is then multiplied elementwise with the input feature to obtain the channel-enhanced feature representation

$$\begin{aligned} F_p &= \text{AvgPool}\left(\text{BN}\left(\text{Conv}_{3\times3}(F)\right)\right) \\ F_{ch} &= F \odot \sigma\left(\text{MLP}\left(F_p\right)\right). \end{aligned} \tag{16}$$

Here, F denotes the input feature map, $\text{Conv}_{3\times3}$ denotes the 3×3 convolution operation, BN denotes batch normalization, AvgPool denotes global average pooling, MLP denotes multilayer perceptron, σ denotes the Sigmoid activation function, and $\odot$ denotes elementwise multiplication.

The spatial branch applies channelwise average pooling and max pooling to the input feature separately and concatenates the two results along the channel dimension to form a spatial attention map $F_s \in \mathbb{R}^{2\times H\times W}$. Subsequently, a 5×5 convolution followed by a Sigmoid function generates the spatial attention weights, which are then multiplied elementwise with the input feature to obtain the spatially enhanced feature representation

$$\begin{aligned} F_s &= \text{Concat}\left[\text{AvgPool}_{\text{chan}}(F),\ \text{MaxPool}_{\text{chan}}(F)\right] \\ F_{sa} &= F \odot \sigma\left(\text{Conv}_{5\times5}(F_s)\right). \end{aligned} \tag{17}$$

Here, $\text{AvgPool}_{\text{chan}}()$ and $\text{MaxPool}_{\text{chan}}()$ denote channelwise average pooling and max pooling, respectively. Concat denotes channel concatenation, $\text{Conv}_{5\times5}$ denotes a 5×5 convolution, σ denotes the Sigmoid activation function, and $\odot$ denotes elementwise multiplication.

The multiscale branch first employs a 1×1 convolution for channel compression, followed by parallel dilated grouped convolutions with dilation rates $r = \{1, 3, 5\}$ for feature extraction. Inspired by hybrid dilated convolution (HDC) [57], this configuration ensures that the selected dilation rates are pairwise coprime, effectively avoiding the "gridding effect" and providing more continuous receptive field coverage. This combination captures both local boundary details ($r = 1$) and broader semantic contextual information ($r = 3, 5$), thereby enhancing the model's robustness to scale variations. The aggregated features are then passed through a 1×1 convolution to restore the channel dimension and fused with the input features

$$\begin{aligned} F' &= \text{GroupConv}_{3\times3,\,\text{dil}=d}\left(\text{Conv}_{1\times1}(F)\right) \\ S &= \sum_{d\in\{1,3,5\}} \text{Conv}_{1\times1}\left(\text{RPReLU}\left(F'\right)\right) \\ F_{ms} &= F \odot \text{Conv}_{1\times1}(S). \end{aligned} \tag{18}$$

Here, $\text{Conv}_{1\times1}$ is a 1×1 convolution, $\text{GroupConv}_{3\times3,\text{dil}=d}$ is a 3×3 grouped convolution with dilation rate $d \in \{1, 3, 5\}$, and RPReLU is the parameterized ReLU; $\sum$ denotes the summation operation for aggregating features with different dilation rates, and $\odot$ denotes elementwise multiplication.

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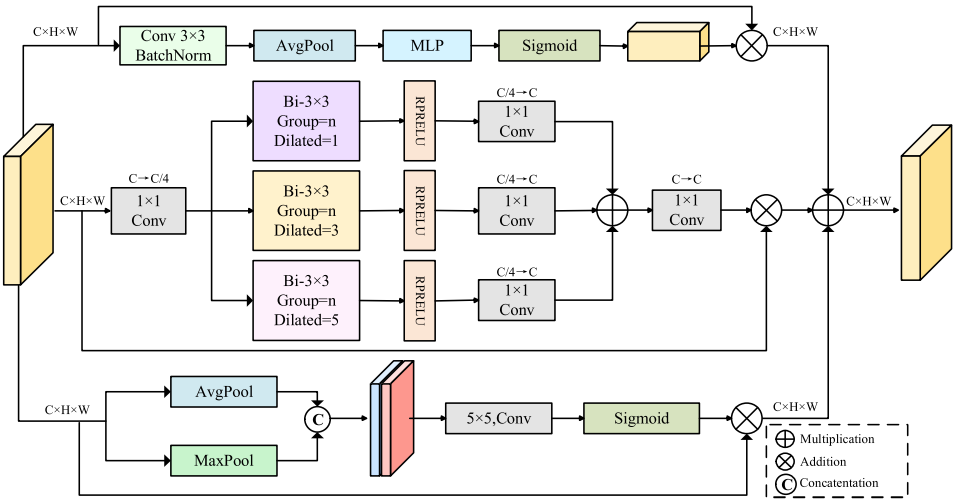


Fig. 5. Schematic of the MFRH model structure.

Finally, the outputs of the three branches are summed to obtain the final output F_{out}

$$F_{\text{out}} = F_{ch} + F_{sa} + F_{ms}. \tag{19}$$

IV. Experiment

A. Experimental Settings

1) Datasets: The ISPRS Vaihingen and Potsdam datasets both provide high-resolution orthophotos of size 6000 × 6000 pixels, annotated with the same six land-cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. For the Vaihingen dataset (33 images in total), we use 15 images (IDs: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 32, 34, 37) for training, image 30 for validation, and the remaining 17 images for testing. For the Potsdam dataset (38 images in total), we use 22 images for training (excluding image 7_10 due to annotation errors), image 2_10 for validation, and 14 images (IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, 7_13) for testing.

The LoveDA dataset is an RSI dataset specifically designed for semantic segmentation tasks in urban, rural, and mixed scenarios. This dataset consists of a total of 5987 images, each cropped to a size of 1024 × 1024 pixels, and annotated with seven categories: buildings, roads, water bodies, vegetation, vehicles, miscellaneous, and background. Among them, 2522 images are used for training, 1669 images for validation, and 1796 images for testing.

2) Implementation Details: The experiments were implemented in PyTorch on an Ubuntu 22.04 system equipped with an NVIDIA RTX 3090 GPU. The model was trained for 70 epochs using the AdamW optimizer with a batch size of 8. The initial learning rate was set to 6×10^{-4} and dynamically adjusted via a cosine annealing strategy. During preprocessing, images from the Vaihingen and Potsdam datasets were cropped into patches of size 512 × 512 pixels. To enhance model robustness, all datasets employed data augmentation strategies, including random flipping and scaling (with scale ratios ranging from 0.5 to 1.5). During inference, test-time augmentation (TTA), combining random flipping and multiscale testing, was adopted to improve the model's generalization capability.

TABLE I
Ablation Experimental Results of MDCNet on the ISPRS Vaihingen Dataset

MGB	SFCM	MFRH	mF1 (%)	mIoU (%)
			91.01	84.16
✓			91.84	85.23
	✓		91.92	85.29
		✓	91.73	85.05
✓	✓	✓	**92.13**	**86.27**

3) Evaluation Metrics: In this experiment, we adopted several key performance metrics commonly used in semantic segmentation of RSIs, including mean intersection over union (mIoU), mean $F1$ score (mF1), and overall accuracy (OA).

mIoU computes the average overlap ratio between predicted and ground-truth pixels across all classes. mF1 is the average of the harmonic means of precision and recall over all classes, providing a balanced evaluation of classification performance. OA represents the proportion of correctly classified pixels among all pixels.

B. Ablation Experiment

We conducted ablation experiments on the MGB module, SFCM module, and MFRH module to verify the necessity of each component and their contributions to performance improvement. All experiments were carried out on the ISPRS Vaihingen dataset, and the results are shown in Table I. When the MGB, SFCM, and MFRH modules were added to the baseline model individually, the mF1 score increased by 0.83%, 0.91%, and 0.72%, respectively, while the mIoU increased by 1.07%, 1.13%, and 0.89%, respectively. When all three modules were added to the baseline model simultaneously, the mF1 score increased by 1.12%, and the mIoU increased by 2.11%. Next, we will provide a more detailed discussion of the effectiveness of these three modules through quantitative and qualitative experimental analysis.

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TABLE II
ABLATION EXPERIMENT RESULTS OF MGB, SFCM, AND MFRH MODULES ON THE ISPRS VAIHINGEN DATASET

Module	Method	mF1(%)	mIoU(%)	OA(%)
MGB	MDCNet	**92.13**	**86.27**	**94.23**
	w/o MGB	91.77	85.12	93.74
	w/o Gate	91.91	85.88	94.07
	CCCC	91.63	84.82	93.71
	LCLC	91.96	85.84	94.03
SFCM	MDCNet	**92.13**	**86.27**	**94.23**
	w/o Local	91.88	85.54	93.85
	w/o Global	91.79	85.29	93.78
	w/o SP	91.96	85.80	94.00
	w/o Haar	91.75	85.26	93.71
	w/o Low	91.86	85.62	93.92
	w/o High	91.94	85.71	93.97
MFRH	MDCNet	**92.13**	**86.27**	**94.23**
	w/o MFRH	91.82	85.41	93.81
	w/o CA+Multi	91.96	85.83	94.03
	w/o SA+CA	91.94	85.72	93.95
	w/o SA+Multi	91.87	85.76	93.98

1) Effectiveness of the MGB Module: The core of the MGB module is an LCCC architecture composed of two types of Transformer blocks connected in series. In particular, the first layer of this architecture utilizes linear attention (denoted as L), while the subsequent three bridging layers all employ condensed attention (denoted as C). Furthermore, the MGB module introduces a scale-aware gating mechanism to reduce the semantic gap before and after multiscale attention. We explored the impact of different MGB module structures on model performance and ultimately selected the optimal configuration.

The corresponding ablation experiment results are shown in Table II. Compared with the model without MGB, introducing MGB improves mIoU and OA by 1.15% and 0.49%, respectively, demonstrating its effectiveness in multiscale feature fusion. Removing the scale-aware gating mechanism further decreases mIoU and mF1 by 0.39% and 0.22%, respectively, confirming the advantage of the adaptive fusion mechanism.

To validate the superiority of the dual-Transformer strategy over a single-Transformer structure, we evaluated a configuration in which all bridging layers employ condensed attention, denoted as CCCC. Compared with the model using only a single Transformer-based bridging structure, the dual-Transformer bridging design improves mIoU from 84.82% to 86.27%, demonstrating the effectiveness and importance of the dual-Transformer design. Furthermore, we compared the LCCC scheme with an alternative configuration that alternates linear attention and condensed attention, referred to as the LCLC scheme, in which the first and third bridging layers adopt linear attention, while the second and fourth bridging layers employ condensed attention. Experimental results show that the LCCC scheme outperforms the LCLC scheme by 0.43% in mIoU, 0.17% in mF1, and 0.20% in OA.

2) Effectiveness of the SFCM Module: To verify the effectiveness of the SFCM dual-domain collaborative mechanism, we conducted a series of ablation experiments, and the experimental results are shown in Table II. When the improved

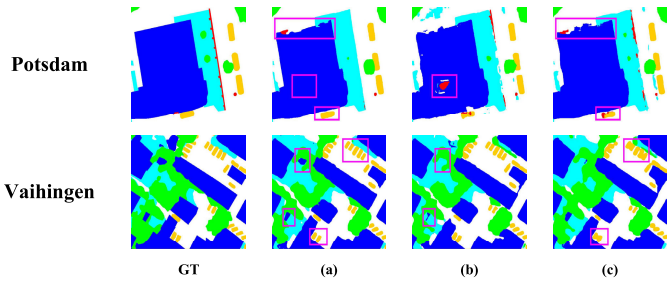


Fig. 6. Visualization analysis of the SFCM dual-domain collaborative mechanism, where magenta boxes highlight the comparison regions worthy of attention. GT denotes the ground truth. (a) SFCM included, (b) spatial-domain branch of SFCM removed, and (c) frequency-domain branch of SFCM removed.

Swin Transformer global branch was removed, the mIoU, mF1, and OA of MDCNet decreased by 0.98%, 0.34%, and 0.45%, respectively, indicating that this global branch can effectively enhance the global contextual modeling capability of the model. Further analysis shows that when only the WLMSA module was introduced into the global branch while SP and large-kernel convolution were removed, the mIoU and mF1 of the model increased by 0.51% and 0.17%, respectively. On this basis, after further introducing SP and large-kernel convolution, the model was able to achieve cross-window information interaction, thereby improving its ability to capture global information and enhancing the efficiency of contextual interaction, leading to a further 0.47% improvement in mIoU. In addition, when the multiscale convolutional local branch was removed, the mIoU, mF1, and OA of the model decreased by 0.73%, 0.25%, and 0.38%, respectively, demonstrating that the local branch plays a crucial role in capturing local detail features. For the Haar frequency-domain branch, completely removing this branch caused the mIoU, mF1, and OA of the model to decrease by 1.01%, 0.38%, and 0.52%, respectively. Further analysis of the contributions of low-frequency and high-frequency features shows that removing only the low-frequency features led to decreases of 0.65%, 0.27%, and 0.31% in mIoU, mF1, and OA, respectively; while removing only the high-frequency features resulted in decreases of 0.56%, 0.19%, and 0.26%, respectively. These results indicate that the precise capture of global and local structures in the spatial domain and the effective supplementation of high- and low-frequency details in the frequency domain achieve complementary advantages between spatial features and spectral information, further improving the overall performance of the model.

Fig. 6 visualizes the effectiveness of the SFCM dual-domain collaborative mechanism on the ISPRS Potsdam and ISPRS Vaihingen datasets, respectively. As shown in Fig. 6(a), the segmentation results generated by MDCNet with SFCM exhibit clear boundaries, semantic consistency, and accurate contours of small objects. This benefits from the SFCM module, which sufficiently integrates semantic contextual information in the spatial domain with structural detail information in the frequency domain. On the contrary, as shown in Fig. 6(b), after removing the spatial-domain branch composed of the global branch and the local branch,

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TABLE III
ABLATION STUDY ON DIFFERENT FREQUENCY DECOMPOSITION METHODS WITHIN THE SFCM MODULE

Transform Method	mIoU(%)	mF1(%)	OA(%)
Fourier Transform	85.00	91.71	93.72
Gabor Wavelet	85.13	91.79	93.87
Haar Wavelet	**86.27**	**92.35**	**94.23**

TABLE IV
ABLATION STUDY OF DIFFERENT MULTISCALE DILATION-RATE BRANCHES ON THE ISPRS VAIHINGEN DATASET

Dilation Rate	mIoU(%)	mF1(%)	OA(%)
{3, 5, 7}	86.08	92.26	94.15
{2, 4, 6}	85.82	92.14	94.06
{1, 3, 5}	**86.27**	**92.35**	**94.23**

the model can still utilize frequency-domain information to capture certain edge and texture variations. However, its spatial contextual modeling capability and local semantic perception capability are significantly degraded, resulting in semantic inconsistency in large homogeneous regions. As shown in Fig. 6(c), after removing the Haar-wavelet-based frequency-domain branch, the model mainly relies on spatial-domain features and lacks the structural prior information provided by high-frequency edges, textures, and gradient variations, thereby causing blurred boundaries and incomplete contours of small objects such as cars.

3) Effectiveness of the MFRH Module: The MFRH module is located at the final layer of the decoder and achieves finer segmentation refinement by fusing shallow and deep features. It consists of three branches: channel attention, multiscale, and spatial attention. As shown in Table II, completely removing the MFRH module leads to a 0.86% drop in mIoU. Moreover, compared to the complete removal of MFRH, retaining any single branch (channel, multiscale, or spatial attention) improves the network's mIoU by 0.35%, 0.31%, and 0.42%, respectively, highlighting the importance of each branch. This fully demonstrates the effectiveness of the MFRH module in enhancing the segmentation accuracy of RSIs.

4) Effectiveness of Different Frequency-Domain Decomposition Methods: To evaluate the rationality of adopting the Haar wavelet in SFCM, we replaced it with other commonly used frequency-domain decomposition methods, and the results are reported in Table III.

The model using FFT achieves the lowest mIoU (85.00%) because mapping features to the global frequency domain leads to a loss of spatial localization information, thereby weakening boundary detection capability. The model using the Gabor wavelet shows a slight improvement (85.13%) by capturing directional texture cues, yet still performs worse than the model using the Haar wavelet (86.27%). The better performance of the Haar wavelet comes from its ability to effectively preserve spatial context, which is critical for semantic segmentation, while extracting frequency-domain information. Moreover, the box-shaped step basis functions of the Haar wavelet are mathematically more suitable than the smooth sinusoidal bases used in FFT or Gabor transforms for representing the sharp edge structures of man-made objects such as buildings and roads.

5) Effectiveness of Different Multiscale Dilation-Rate Branches: We are inspired by HDC [57] and adopt dilation-rate combinations with no common divisor greater than 1 to mitigate the gridding effect. To evaluate the rationality of the dilation-rate design in MFRH, we compare three typical dilation-rate combinations on the ISPRS Vaihingen dataset: $\{3, 5, 7\}$, $\{2, 4, 6\}$, and the adopted $\{1, 3, 5\}$. Quantitative results are reported in Table IV.

As shown in the experimental results, the proposed dilation rate combination $\{1, 3, 5\}$ achieves the best performance, attaining an mIoU of 86.27%, outperforming $\{2, 4, 6\}$ (85.82%) and $\{3, 5, 7\}$ (86.08%). The inferior result of $\{2, 4, 6\}$ indicates that a common divisor among the dilation rates introduces gridding artifacts and weakens feature aggregation. Although $\{3, 5, 7\}$ avoids this issue, its larger dilation rates lead to sparser sampling, which may miss fine boundary details. Our designed $\{1, 3, 5\}$ configuration includes a nondilated branch, which better balances local detail preservation and large-range context modeling.

TABLE V
SENSITIVITY ANALYSIS OF ATTENTION HEAD COUNT h IN SFCM ON ISPRS VAIHINGEN

h	mIoU (%)	mF1 (%)	OA (%)	FPS
1	85.32	91.46	93.45	21.68
2	85.63	91.68	93.71	22.16
4	86.04	92.02	94.04	22.65
8	**86.27**	**92.25**	**94.23**	**22.93**
16	86.12	92.17	94.14	22.84

6) Effectiveness of Multihead Configuration and Sensitivity Analysis: To investigate the model's sensitivity to key hyperparameters and further validate the effectiveness of the multihead design, we conduct an ablation study on the number of cross-attention heads h in the SFCM on the ISPRS Vaihingen dataset.

As shown in Table V, model performance is strongly correlated with the number of attention heads. When h is set to 1 or 2, the network exhibits limited capacity to model complex cross-domain interactions, yielding suboptimal results; for instance, an mIoU of 85.32% at $h = 1$. As h increases to 8, all metrics steadily improve, achieving a peak mIoU of 86.27%, an OA of 94.23%, and the highest inference speed of 22.93 FPS. This indicates that projecting spatial and frequency features into multiple subspaces enables the network to capture complementary information more effectively. However, further increasing h to 16 leads to a slight performance drop, with mIoU decreasing to 86.12% and a reduction in FPS. This degradation may stem from redundant feature representations and computational fragmentation introduced by excessive attention heads, which complicate optimization. Therefore, based on empirical evaluation, we adopt $h = 8$ as the default configuration in MDCNet.

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TABLE VI
COMPARISON WITH THE SOTA METHODS ON THE POTSDAM DATASET

Method	Imp.Surf(%)	Building(%)	Lowveg(%)	Tree(%)	Car(%)	mF1(%)	mIoU(%)	OA(%)
A2-FPN	93.83	96.28	87.26	88.23	95.74	92.27	85.88	91.17
ABCNet	92.88	95.49	86.83	88.59	95.01	91.76	84.97	90.41
MANet	92.42	95.06	86.01	88.02	95.84	91.47	84.51	89.85
BANet	93.19	96.01	87.54	88.87	95.17	92.15	85.63	90.93
TransUNet	92.36	95.45	86.63	87.95	95.23	91.52	84.93	90.52
DC-Swin	94.38	97.04	88.12	89.19	96.11	92.96	87.07	91.85
UNetFormer	93.52	95.97	87.55	88.51	95.86	92.28	85.87	91.05
ConvLSRNet	94.37	97.02	87.85	89.67	96.22	93.03	87.33	91.71
ER-Swin	93.72	**97.13**	88.08	89.64	96.53	93.04	87.20	91.58
RTMamba	93.91	96.52	**88.98**	89.98	96.64	92.86	87.15	91.56
DOCNet	94.02	96.59	87.55	89.04	95.97	92.63	86.67	91.34
MDCNet	**94.63**	97.06	88.27	**90.78**	**96.79**	**93.51**	**87.61**	**92.08**

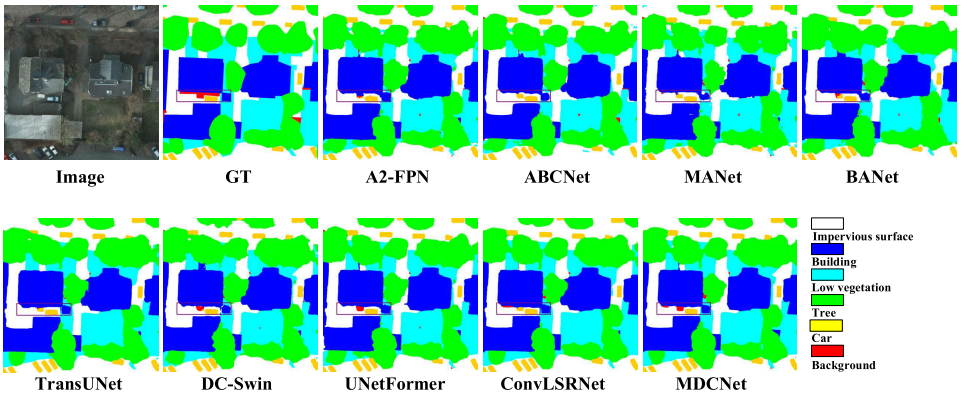


Fig. 7. Qualitative comparison with state-of-the-art methods on the ISPRS Potsdam dataset.

C. Compared Experiment

We compared the proposed MDCNet with representative state-of-the-art semantic image segmentation methods, including ABCNet [58], A2-FPN [59], MANet [60], BANet [61], TransUNet [62], DC-Swin [63], UNetFormer [64], ConvLSRNet [65], ER-Swin [66], RTMamba [41], DOCNet [18], and RS3Mamba [40].

To ensure fairness and rigor in the comparison, we explicitly clarified the training and inference settings for these baseline methods. In particular, ABCNet, A2-FPN, MANet, BANet, TransUNet, DC-Swin, UNetFormer, ConvLSRNet, and DOCNet were retrained under the same training configuration as MDCNet to obtain experimental results on each dataset. For the remaining three methods, namely ER-Swin, RTMamba, and RS3Mamba, we directly adopted the quantitative results reported in their original papers under identical dataset splits and evaluation metrics. Furthermore, all retrained baseline methods employed the same TTA strategy as MDCNet during inference, thereby eliminating performance bias caused by differing experimental setups and providing more convincing validation of the effectiveness of the proposed method.

1) Results on the ISPRS Potsdam Dataset: We conducted a comprehensive evaluation of MDCNet on the ISPRS Potsdam dataset. As shown in Table VI, MDCNet achieved the highest OA of 92.08%, surpassing Swin Transformer-based methods such as ER-Swin (91.58%) and DC-Swin (91.85%). Meanwhile, its mF1 reached 93.51%, outperforming baseline models such as ConvLSRNet and RTMamba, which demonstrates a more balanced segmentation capability. Notably, MDCNet achieved an $F1$ score of 96.79% in the highly challenging "car" category. In addition, its mIoU reached the best value of 87.61%, exceeding advanced methods such as ConvLSRNet (87.33%). These results indicate that MDCNet can predict segmentation boundaries more accurately and effectively handle overlapping regions among different categories, thereby improving the overall segmentation performance in complex scenes.

Fig. 7 presents the segmentation visualization results of different methods on the Potsdam dataset. Compared with other methods, MDCNet performs better in boundary delineation and small-object segmentation, effectively reducing misclassification and artifacts. In complex transition regions of the "buildings" category, MDCNet achieves more accurate boundary localization and produces clearer segmentation results than methods such as A2-FPN and MANet. Although Transformer-based models such as DC-Swin, UNetFormer, and ConvLSRNet have advantages in global contextual modeling, they still show certain limitations in representing fine-grained details, especially in dense vegetation regions where unnatural category transitions are likely to occur. On the contrary, MDCNet effectively integrates spatial-domain and frequency-domain information through its dual-domain fusion mechanism, enabling better preservation of fine details, clearer boundaries, and smoother category transitions. The highlighted regions in Fig. 7 further verify the robustness of MDCNet for segmentation in complex scenes.

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TABLE VII
COMPARISON WITH THE SOTA METHODS ON THE ISPRS VAIHINGEN DATASET

Method	Imp.Surf(%)	Building(%)	Lowveg(%)	Tree(%)	Car(%)	mF1(%)	mIoU(%)	OA(%)
A2-FPN	96.66	95.34	84.46	90.06	86.97	90.71	83.32	93.12
ABCNet	96.68	94.99	84.43	90.26	86.26	90.52	83.04	93.06
MANet	96.22	93.02	83.69	89.76	85.53	89.64	81.56	92.25
BANet	96.53	94.93	84.15	90.01	87.62	90.64	83.21	92.93
TransUNet	96.31	93.28	84.01	89.53	86.37	89.91	82.42	92.69
DC-Swin	97.16	96.01	85.41	90.38	88.53	91.51	84.64	93.69
UNetFormer	96.88	95.41	84.44	90.05	86.71	90.71	83.34	93.27
ConvLSRNet	97.16	96.15	85.37	90.47	90.12	91.85	85.23	93.77
ER-Swin	94.05	**97.07**	85.75	90.61	90.95	91.69	84.89	92.36
RTMamba	93.75	95.30	**85.97**	90.93	88.70	91.08	83.92	91.30
DOCNet	96.87	96.06	85.05	90.02	88.82	91.38	84.54	93.47
MDCNet	**97.18**	96.41	85.64	**90.97**	**91.03**	**92.25**	**86.27**	**94.23**

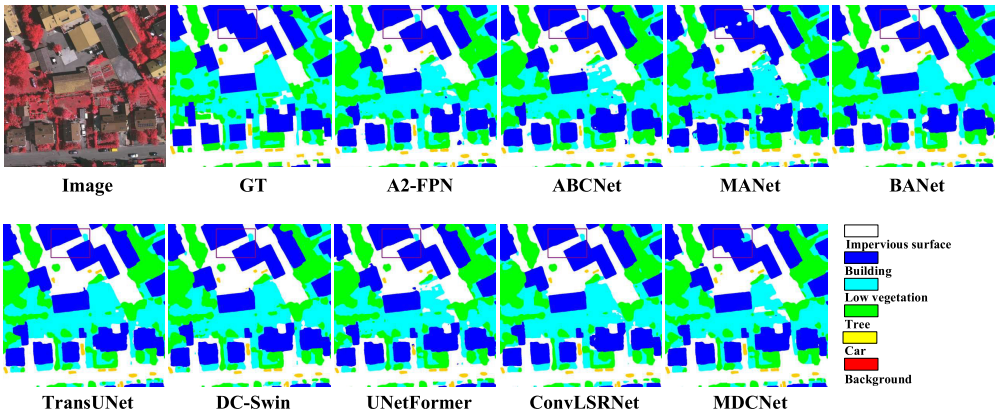


Fig. 8. Qualitative comparison with state-of-the-art methods on the ISPRS Vaihingen dataset.

2) Results on the ISPRS Vaihingen Dataset: Table VII presents the experimental results on the ISPRS Vaihingen dataset. MDCNet achieves the highest scores on all three core metrics, namely mF1, mIoU, and OA. Compared with the second-best method, ConvLSRNet, MDCNet improves mF1 by 0.40%, mIoU by 1.04%, and OA by 0.46%. These performance gains are mainly attributed to the MGB and SFCM modules, which enhance the network's ability to focus on key features and enable more accurate discrimination among different land-cover categories. Notably, MDCNet performs particularly well in small-object segmentation, achieving an $F1$ score of 91.03% for the "car" category. This indicates that MDCNet has strong capability in identifying small objects and complex boundary regions.

A2-FPN achieves competitive $F1$ scores across all categories by leveraging spatial and channel attention mechanisms. ER-Swin, benefiting from the advantage of its diagonal interactive attention module in global information extraction, performs particularly well in the "building" category, achieving an $F1$ score of 97.07%. In addition, ConvLSRNet balances local and global modeling through an efficient Transformer module, obtaining an mF1 of 91.85% and an mIoU of 85.23%. However, although these methods perform well in global contextual modeling, they still exhibit limitations in fine-grained feature representation for small objects such as cars. On the contrary, MDCNet achieves the best $F1$ scores on challenging categories such as "tree" (90.97%) and "car" (91.03%), which fully validates its excellent adaptability to fine-grained structures and complex remote sensing scenes.

To provide a more intuitive comparison of the segmentation performance of different algorithms, Fig. 8 presents the visualization results of each method on the ISPRS Vaihingen dataset. Limited by their lightweight attention mechanisms, ABCNet and MANet exhibit obvious loss of fine details in their segmentation results. Although Transformer-based methods such as UNetFormer and ConvLSRNet perform better in global contextual modeling, they are prone to semantic overlap between adjacent objects in detail-rich regions, resulting in blurred boundaries and inaccurate localization. On the contrary, MDCNet demonstrates the best overall segmentation performance. Whether for large-scale objects with irregular boundaries, such as buildings, or for small and densely distributed vehicles, MDCNet is able to generate complete and accurate segmentation results, precisely locate the boundaries of densely packed targets, and effectively avoid adhesion and missegmentation.

3) Results on the LoveDA Dataset: Compared with the urban-focused ISPRS Potsdam and Vaihingen datasets, LoveDA poses greater challenges due to severe class imbalance and more diverse urban–rural landscapes. Table VIII presents the performance comparison results on this dataset. MDCNet achieves the highest mIoU of 53.02%, outperforming DOCNet (52.54%) and DC-Swin (52.26%), which indicates that MDCNet has strong generalization ability and efficient

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TABLE VIII
COMPARISON WITH THE SOTA METHODS ON THE LOVEDA DATASET

Methods	Background(%)	Building(%)	Road(%)	Water(%)	Barren(%)	Forest(%)	Agriculture(%)	mIoU(%)
A2-FPN	42.84	56.08	52.21	74.76	13.90	43.91	55.34	48.43
ABCNet	41.51	51.93	50.38	78.56	10.67	44.21	58.79	48.01
MANet	42.16	55.65	54.77	76.44	14.86	44.64	54.34	48.98
BANet	42.26	56.09	52.50	73.99	14.91	46.00	52.30	48.29
TransUNet	44.42	58.71	56.41	**81.23**	17.80	48.29	59.49	52.33
DC-Swin	43.49	59.67	57.19	79.61	17.12	47.55	61.15	52.26
UNetFormer	42.57	55.58	54.40	78.15	16.09	43.61	53.21	49.09
ConvLSRNet	43.30	60.10	52.65	80.44	18.90	47.44	57.86	51.53
RS3Mamba	39.72	58.75	57.92	61.00	**37.24**	39.67	33.98	50.93
RTMamba	44.72	56.43	52.49	79.90	20.96	44.46	55.06	50.57
DOCNet	**53.51**	**61.67**	55.37	70.65	32.08	41.95	58.74	52.54
MDCNet	44.98	61.16	**57.95**	80.45	18.24	**48.42**	**61.22**	**53.02**

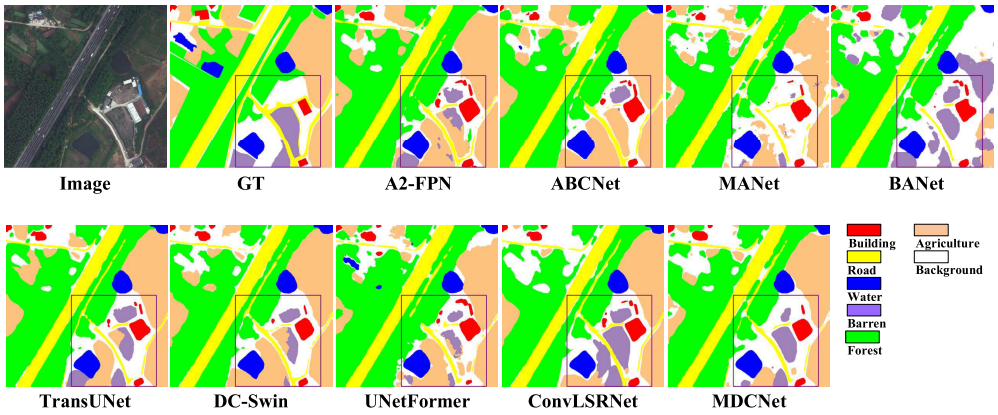


Fig. 9. Qualitative comparison with state-of-the-art methods on the LoveDA dataset.

segmentation capability in complex and diverse land-cover scenes.

From the perspective of IoU scores for each category, MDCNet performs particularly well in the "forest" (48.42%), "road" (57.95%), and "farmland" (61.22%) categories, demonstrating its precise recognition ability for fine-grained targets and large-scale features. It is worth noting that although DOCNet and TransUNet have slight advantages in the "building" (61.67%) and "water" (81.23%) categories, respectively, MDCNet also maintains high competitiveness in these categories, achieving IoU scores of 61.16% and 80.45%, respectively. Overall, despite some tradeoffs in individual categories, MDCNet still leads DOCNet and other mainstream models with the highest mIoU score of 53.02%.

Fig. 9 presents the visualization results of MDCNet on the LoveDA dataset. Compared with other mainstream models, MDCNet achieves better segmentation quality, especially in terms of detail preservation and complex boundary delineation. For regions dominated by forest and road, its predictions are highly consistent with the ground truth, with clearer boundaries and fewer missegmented areas. Compared with UNetFormer and ConvLSRNet, MDCNet also performs better on challenging classes such as wasteland and farmland. In wasteland areas, MDCNet is able to capture smoother and more continuous boundaries, effectively reducing oversegmentation and boundary blurring. These visual results further indicate that MDCNet has stronger boundary alignment ability and better interclass discrimination ability.

TABLE IX
COMPLEXITY ANALYSIS CONDUCTED ON A SINGLE NVIDIA GEFORCE RTX 3090 GPU WITH AN INPUT SIZE OF 512 × 512. MIOU AND MF1 ARE REPORTED ON THE ISPRS VAIHINGEN DATASET

Model	Backbone	FLOPs (G)	Params (M)	FPS	mIoU (%)	mF1 (%)
UNetformer	CNNs	**23.48**	**11.69**	**65.14**	83.34	90.71
TransUNet	Transformer	258.20	105.32	18.74	82.42	89.91
DCSwin	Transformer	140.30	66.90	31.78	84.64	91.51
ConvLSRNet	CNN+Transformer	141.76	68.04	26.32	85.23	91.85
MDCNet (Ours)	Transformer	274.61	110.20	22.93	**86.27**	**92.25**

D. Model Complexity Analysis

To evaluate the computational cost of MDCNet, we analyzed the computational complexity (FLOPs) and parameter count (Params) of different methods. All models were tested on a single NVIDIA GeForce RTX 3090 GPU with an input resolution of 512 × 512. The quantitative results are summarized in Table IX.

UNetFormer [64], based on a lightweight CNN backbone network, maintains a low computational cost, with only 23.48G FLOPs, making it suitable for resource-constrained application scenarios. However, due to its limited capability in capturing complex semantic information, its segmentation accuracy is relatively restricted, achieving an mIoU of only 83.34%. On the contrary, Transformer-based methods generally possess stronger global contextual modeling capability, but they often require higher computational resources. Although DC-Swin [63] effectively reduces the computational cost by leveraging the window-based attention mechanism

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of the Swin Transformer, with 66.90 M parameters and 140.30 G FLOPs, respectively, its segmentation accuracy falls significantly behind that of MDCNet, trailing by 1.63% in terms of the mIoU metric. In addition, compared with the recent advanced method ConvLSRNet [65], MDCNet further improves the mIoU by 1.04% and the mF1 by 0.40%. It is worth noting that, compared with TransUNet [62], which also has a large parameter scale, MDCNet introduces less than 5% additional parameters while achieving a 3.85% improvement in mIoU. This indicates that the performance gain of MDCNet does not simply rely on parameter stacking, but mainly benefits from a more effective architectural design. Meanwhile, MDCNet still achieves an inference speed of 22.93 FPS on a single RTX 3090 GPU, demonstrating that it maintains practically valuable inference efficiency while pursuing high segmentation accuracy.

V. Discussion

A. Limitations

Although MDCNet effectively improves the segmentation accuracy of RSIs, its parameter count of up to 110.20 M results in a relatively large model size. On resource-constrained edge devices, such as unmanned aerial vehicles or onboard satellite processing units, this poses significant challenges to the practical deployment of the model. In addition to computational resource limitations, another limitation of MDCNet lies in its reliance on single-modal optical data. Since only RGB image information is utilized, the model struggles to fully address the common "same spectrum, different objects" problem in remote sensing scenarios. For example, in the absence of geometric height information provided by a DSM, concrete roofs and paved roads may exhibit highly similar spectral characteristics, making it difficult for the model to distinguish between them. Furthermore, single-modal RGB models are relatively sensitive to imaging conditions. Factors such as shadows and illumination variations can significantly alter the spectral responses of ground objects, increase intraclass variance, and weaken critical texture details, thereby affecting the robustness of the model in complex environments.

B. Future Work

To address the limited edge deployment caused by the large number of model parameters, we plan to introduce a knowledge distillation strategy in future work. In particular, the dual-domain feature representations learned by MDCNet will be transferred to a lightweight CNN or Mamba-based student model, thereby reducing the number of parameters and computational overhead while maintaining high segmentation accuracy. In addition, to overcome the limitations of single-modal optical data, future work will further extend MDCNet into a multimodal fusion architecture. In particular, we will introduce the DSM as an additional input and deeply fuse the geometric information it provides with RGB texture information, so as to better distinguish objects with similar spectral characteristics but different physical structures. Finally, we will explore the use of diffusion models to refine the boundaries of complex objects and generate highly challenging synthetic remote sensing scenes. This generative data augmentation strategy will effectively strengthen the feature representation capability of the model and further improve its generalization performance in complex remote sensing environments.

VI. Conclusion

This article proposes an efficient network architecture, MDCNet, specifically designed for semantic segmentation of RSIs. The network innovatively introduces the MGB module, which can adaptively perform deep interaction with multiscale features, thereby effectively capturing object information at different spatial scales. Meanwhile, the SFCM module extracts and fuses features from both the frequency and spatial domains, alleviating the segmentation challenges caused by drastic grayscale variations and enhancing the ability to recognize fine-grained structures. In addition, the MFRH module combines spatial/channel attention with multiscale atrous convolution mechanisms, fully aggregating global contextual information during the decoding stage and further improving segmentation accuracy in complex scenes. Extensive experiments on the ISPRS Potsdam, ISPRS Vaihingen, and LoveDA datasets demonstrate that MDCNet significantly outperforms existing mainstream methods across multiple evaluation metrics.

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Hailong Ma received the B.Eng. degree in computer science and technology from the School of Information Engineering, North China University of Water Resources and Electric Power, Zhengzhou, China, in 2023. He is currently pursuing the master's degree in computer technology with the School of Computer Science (National Pilot Software Engineering School), Beijing University of Posts and Telecommunications, Beijing, China.

His research interests include computer vision and semantic segmentation of RSIs.



Songwei Pei (Member, IEEE) received the Ph.D. degree in computer science from the Institute of Computing Technology (ICT), Chinese Academy of Sciences (CAS), Beijing, China, in 2011.

He is currently an Associate Professor with the School of Computer Science (National Pilot Software Engineering School), Beijing University of Posts and Telecommunications, Beijing. He has published several dozen technical articles in renowned international journals and conferences, including the IEEE/ACM Transactions series, Elsevier, and Springer journals. His current research interests include computer vision, deep learning, computer architecture, and VLSI design.

Dr. Pei is a member of several academic and specialized committees. He serves as a reviewer for several renowned journals and as a program committee member for significant conferences.



Bingfeng Liu received the B.Eng. degree in data science and big data technologies from the School of Computer Science and Technology, Chinese University of Mining and Technology, Xuzhou, China, in 2024. He is currently pursuing the Ph.D. degree in computer science with the School of Computer Science (National Pilot Software Engineering School), Beijing University of Posts and Telecommunications, Beijing, China.

His research interests include image fusion and stereo matching.



Qian Li (Member, IEEE) is currently an Assistant Professor at Beijing University of Posts and Telecommunications (BUPT), Beijing, China. She has authored more than 30 articles published in top-tier conferences and journals such as NeurIPS, WWW, ACM MM, and AAAI, amassing more than 2000 citations on Google Scholar. Her research interests focus on natural language processing, multimodal learning, and knowledge graphs.

Dr. Li received the Best Paper Nomination at CIKM 2022, two ESI Highly Cited Papers, the IEEE TCCLD 2024 Technical Innovation Award, the CodaLab championship, and the First Prize for Scientific and Technological Progress from the State Grid Big Data Center. She has also served as the Area Chair for ACL and NeurIPS, a Senior Program Committee Member for IJCAI, and a Program Committee Member for IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, ICML, NeurIPS, and ICLR.



Shangguang Wang (Senior Member, IEEE) received the Ph.D. degree from Beijing University of Posts and Telecommunications, Beijing, China, in 2011.

He is a Professor with the School of Computer Science (National Pilot School of Software Engineering), Beijing University of Posts and Telecommunications. He has published more than 150 articles. His research interests include service computing, mobile edge computing, and satellite computing.

Dr. Wang is a fellow of IET. He is currently serving as the Chair for the IEEE Technical Committee on Services Computing and the Vice-Chair for the IEEE Technical Committee on Cloud Computing. He has also served as general chairs or program chairs of more than ten IEEE conferences, and associate editors of several journals, such as IEEE TRANSACTIONS ON SERVICES COMPUTING and *Journal of Software: Practice and Experience*.

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